Practice workbook · Finding stationary points

Setting dy/dx = 0 and solving (quadratics, factorised cubics or trig equations) to find the coordinates of stationary points.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-CAL-08

1. Question 7 · part a · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part a only.

7 The curve C has equation $y = \dfrac{x}{x^2 + 4}$

(a) Using calculus, find the coordinates of the stationary points on C.

(5)

(b) Show that $\dfrac{d^2y}{dx^2} = \dfrac{2x(x^2 - 12)}{(x^2 + 4)^3}$

(4)

(c) Hence, or otherwise, determine the nature of each of these stationary points.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 7 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 9 · part b.ii · 7 marks

2019 JUN · Paper 02

Further practice · Attempt part b.ii only.

9 The curve C, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that C passes through the origin.

(4)

(b) (i) Show that C has a minimum point at $x = 2$ and a maximum point at $x = 1$

(ii) Find the exact value of the y coordinate at each of these points.

(7)

The curve has another turning point at A.

(c) (i) Find the coordinates of A.

(ii) Determine the nature of this turning point.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 9 · part b.ii · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 9 · part c.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part c.i only.

9 The curve C, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that C passes through the origin.

(4)

(b) (i) Show that C has a minimum point at $x = 2$ and a maximum point at $x = 1$

(ii) Find the exact value of the y coordinate at each of these points.

(7)

The curve has another turning point at A.

(c) (i) Find the coordinates of A.

(ii) Determine the nature of this turning point.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 9 · part c.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 9 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

9 A curve C has equation

$$y = \frac{2 + 4x - x^2}{2x + 1} \quad x \neq -\frac{1}{2}$$

(a) Write the equation of C in the form $ax^2 + (by - 4)x + (y - c) = 0$, where a, b and c are integers whose values are to be found.

(3)

(b) Hence show that x is real when $y \leqslant 2$ and when $y \geqslant 3$

(4)

(c) Find the coordinates of the stationary points on C.

(6)

(d) Sketch C showing clearly

(i) the exact coordinates of the points where C crosses the x-axis,

(ii) the asymptote to C that is parallel to the y-axis,

(iii) the coordinates of the stationary points.

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 6 2 2 8 5 A 0 2 4 3 2

4. Question 9 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 11 · part c · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

11 (a) Using a formula on page 2 show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Hence show that $\left(2\cos^2 A - 1\right)^2 = \dfrac{\cos 4A + 1}{2}$

(3)

The curve with equation $y = \dfrac{\sin 2x}{2} + \dfrac{\left(2\cos^2 x - 1\right)^2}{2} + \dfrac{1}{8}$ has a stationary point P in the range $0 \leqslant x \leqslant \dfrac{\pi}{6}$

(c) Find the exact coordinates of P

(7)

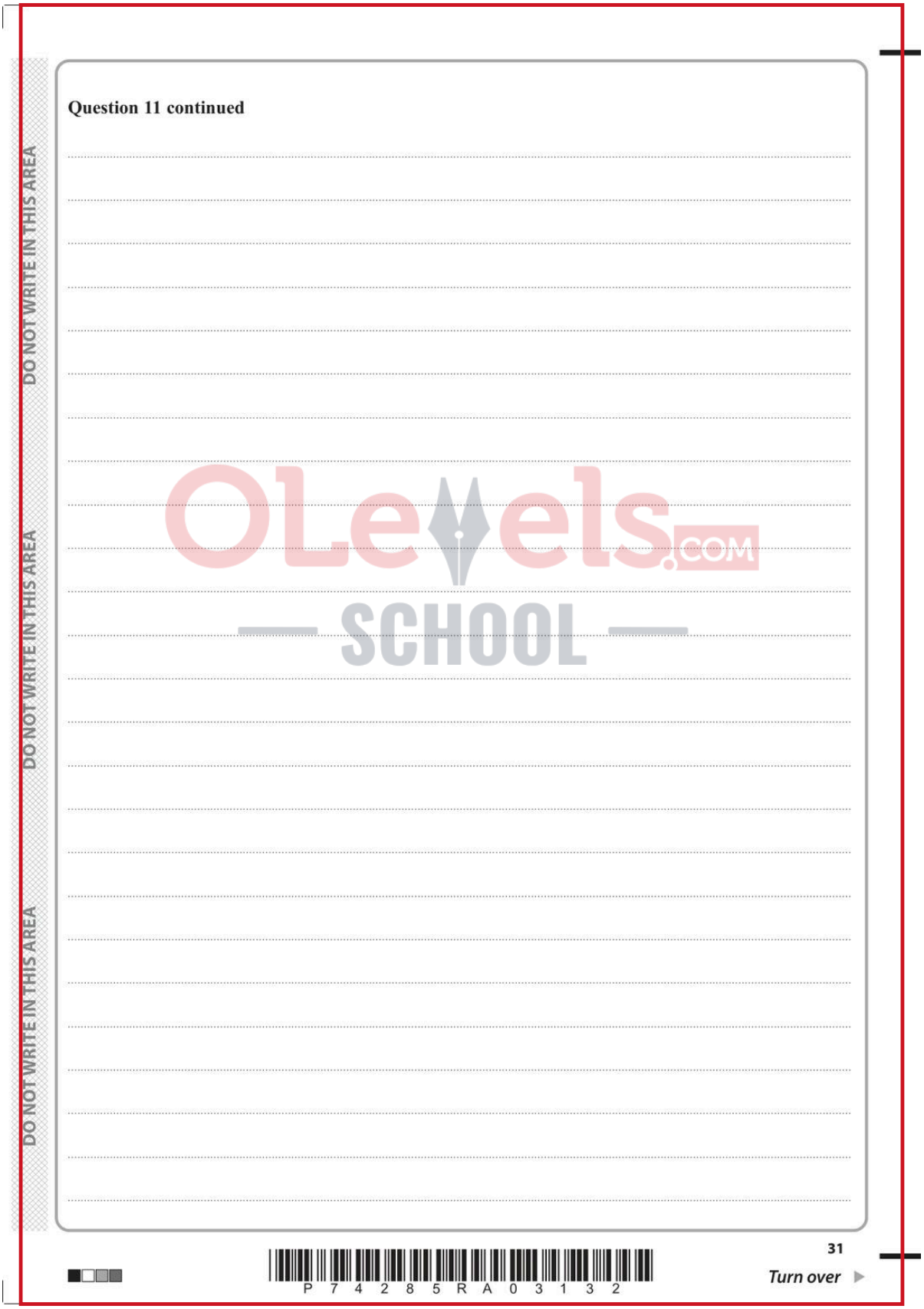
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 11 · part c · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.



5. Question 11 · part c · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 11 is 12 marks)

5. Question 11 · part c · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 7 · part b.i · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

7 The point with coordinates $(4, -104)$ lies on the curve C with equation $y = f(x)$

Given that $f'(x) = 4x^3 - 12x^2 - 19x + 12$

(a) (i) show that C passes through the origin,

(4)

(ii) show that C has a maximum at the point on the curve where $x = 0.5$

(3)

The curve C has another turning point at A and another turning point at B.

Given that the x coordinate of A is negative,

(b) (i) find the coordinates of A and the coordinates of B,

(5)

(ii) determine the nature of these turning points.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

20

P 7 1 6 6 1 A 0 2 0 3 2

6. Question 7 · part b.i · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 10 · part b · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

10

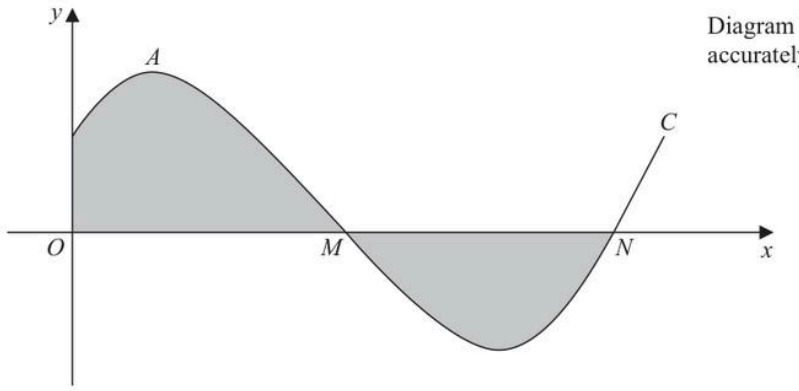


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows the curve C with equation $y = \frac{1}{2} + \sin 3x$ where $0 \leqslant x < \frac{2\pi}{3}$

The curve C crosses the x-axis at the points M and N

(a) Show that the coordinates of M are $\left(\frac{7\pi}{18}, 0\right)$ and find the coordinates of N

(3)

The curve C has a maximum at the point A

(b) Find the coordinates of A

(4)

(c) Find an equation of the tangent to C at M

Give your answer in the form $ay + b\sqrt{3}x - c\sqrt{3}\pi = 0$ where a, b and c are integers to be found.

(4)

The finite region, shown shaded in Figure 4, is bounded by the curve C, the y-axis and the part of the x-axis from O to N

(d) Use algebraic integration to find, to 3 significant figures, the total area of the shaded region.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

7. Question 10 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 4 · part a · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

4 The curve C has equation

$$y = x^3 - 3x^2 - 24x + 6$$

(a) Use calculus to find the coordinates of each of the stationary points on C.

(4)

(b) Determine the nature of each of these stationary points.
Justify your answers.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 4 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL